Tax-Aware Rebalancing and TLH: Executive Summary

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Bottom line. Across 192 matched TLH-versus-no-TLH comparisons, harvesting reliably cut taxes (in 86% of runs, about +$10,182 per $1M on average), but its net effect on portfolio value is decided by how well the wash-sale replacement ETF tracks the original. Net of tax, tracking, and cost, harvesting helped in 55% of runs pre-liquidation and 37% after liquidation.

What we built

A transparent, lot-level, tax-aware simulation engine (short/long-term classification, $3k ordinary offset, character-preserving carry-forward, 30-day wash-sale rule with automatic proxy substitution, and calendar plus drift-band rebalancing) wrapped in an interactive dashboard, plus a 384-run backtest and a ranked strategy playbook. Every reported number is reproducible from the committed code and data.

How we tested it

Four portfolios (40/60 and 100/0 model ETFs, a 2-ETF and a 3-ETF mix) across six market periods, eight rebalancing strategies, each run with and without harvesting: 192 matched comparisons. All runs use the same engine, $1M of capital, 35% short / 20% long-term tax rates, 12 bps round-trip cost, and a 10% harvest threshold, so the only difference within a pair is harvesting.

Headline result: decompose the value-add

For each portfolio, the net after-tax value-add of harvesting splits into tax saved, the tracking difference of the replacement ETFs, and extra cost. Tax saved is positive everywhere; replacement tracking decides the net.

Portfolio	Tax saved	Replacement tracking	Extra cost	Net value-add	Avg TLH events
3-ETF (SPY/TLT/GLD)	+$17,973	+$15,371	-$3,009	+$30,335	19
2-ETF (IVV/EFA)	+$8,860	+$9,277	-$2,259	+$15,879	11
40/60 (TA ETF)	+$8,524	-$18,761	-$3,038	-$13,275	97

100/0 (TA ETF)	+$5,370	-$28,588	-$3,272	-$26,490	56

Worst run: **100/0 (TA ETF)**, past 20 years, at -$492,226, of which -$410,475 was replacement tracking (a growth sleeve swapped into a broad S&P 100 fund) and -$73,185 was tax. Best run: **3-ETF (SPY/TLT/GLD)**, past 20 years, at +$210,784, also tracking-driven (the replacement outperformed by +$184,832). Single-run extremes reflect replacement tracking in both directions; the reliable effect is the tax saving.

Recommendations

- Curate the replacement-ETF map to closely tracking, style-matched substitutes. It is the biggest lever.
- Enable TLH for diversified portfolios with good replacements; confirm tracking before enabling it for concentrated or style-tilted portfolios.
- Default to infrequent, wide-band rebalancing (yearly, relative-25%, absolute-20%); monthly rebalancing ranks worst overall.
- Communicate value using post-liquidation figures, since much of the tax benefit is deferral.

Key assumptions

Price-basis returns for comparability; Sharpe defined as CAGR over volatility (Rf = 0); active-ticker universe (some survivorship bias); federal capital gains only; long-only. The interactive app can additionally model dividends and reinvestment.